

Sampling Rate Analysis

C6:5 Engine Specs

1:10 Nike Smoke Rocket
Rocket weight = 68.04 g

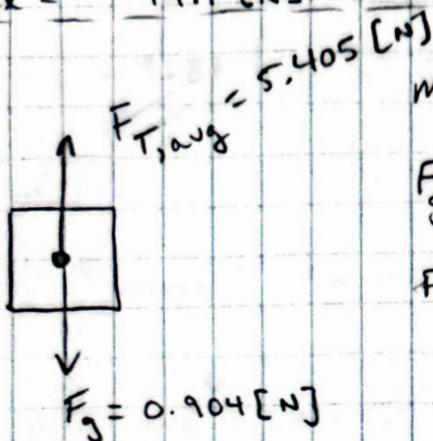
impulse: 10 N·s

Burn time: 1.85 s

weight = 24.1 g

Thrust max = 14.1 [N] → max $a \approx 15g$

EBD



$$m_{TOTAL} = 0.09214 \text{ [kg]}$$

$$F_g = m_{TOTAL} \cdot a = 0.09214 \text{ [kg]} \cdot 9.81 \text{ [m/s}^2]$$

$$F_g = 0.904 \text{ [N]}$$

$$\text{impulse} = F_{T,avg} \cdot t_{burn}$$

$$10 \text{ [N·s]} = F_{T,avg} \cdot 1.85 \text{ [s]}$$

$$\therefore F_{T,avg} = 5.405 \text{ [N]}$$

$$F_{avg} = 5.405 \text{ [N]} - 0.904 \text{ [N]}$$

$$F_{avg} = 4.501 \text{ [N]}$$

Find a_{avg} for rocket flight: $F_{avg} = m \cdot a_{avg} = 0.09214 \text{ [kg]} \cdot a_{avg}$

$$\therefore a_{avg} = 48.85 \text{ [m/s}^2]$$

From C6:5 thrust curve: $F_{thrust} = F_g @ \approx 1.84 \text{ sec}$

$$v_{max} \approx a_{avg} \cdot t = 48.85 \text{ [m/s}^2] \cdot 1.84 \text{ [s]}$$

$$v_{max} \approx 89.88 \text{ [m/s]}$$

Sensor Sampling Rate:

50 Hz → $\Delta t = 0.02 \text{ [s]} \rightarrow \Delta x = 89.88 [0.02] \rightarrow \Delta x = 1.79 \text{ [m]}$

100 Hz → $\Delta t = 0.01 \text{ [s]} \rightarrow \Delta x = 89.88 [0.01] \rightarrow \Delta x = 0.899 \text{ [m]}$

Taylor Law
①

Sensor Sampling Rate cont:

$$\textcircled{a} 45 \text{ Hz} \rightarrow \Delta t = 0.022 \text{ [s]} \rightarrow \Delta x / \text{sample} = 89.88 [0.22] \approx 2 \text{ m}$$

Apogee Estimation:

@ apogee

$$t_{\text{total}} = t_{\text{preflight}} + t_{\text{burn}} + t_{\text{drift}} + t_{\text{descent}}$$

$$t_{\text{preflight}} \approx 10 \text{ [s]}, \quad t_{\text{burn}} = 1.85 \text{ [s]}$$

Find t_{drift} :

$$V_{\text{apogee}}^0 = V_{\text{burnout}} + a t_{\text{drift}}, \quad a = -9.81 \text{ [m/s}^2\text{]}$$

$$V_{\text{burnout}} = 89.88 \text{ [m/s]}$$

$$0 = 89.88 \text{ [m/s]} - 9.81 \text{ [m/s}^2\text{]} \cdot t_{\text{drift}} \rightarrow \therefore t_{\text{drift}} \approx 9.16 \text{ [s]}$$

Find t_{descent} :

$$\text{Find } X_{\text{apogee}} = X_{\text{burnout}} + V_{\text{burnout}} \cdot t_{\text{drift}} + \frac{1}{2} a t_{\text{drift}}^2$$

$$X_{\text{burnout}} = X_0^0 + Y_0^0 + \frac{1}{2} a t^2 = \frac{1}{2} (a_{\text{avg}}) t_{\text{burn}}^2$$

$$X_{\text{burnout}} = 48.85 \cdot (1.85)^2 \rightarrow X_{\text{burnout}} = 167.189 \text{ [m]}$$

$$X_{\text{apogee}} = 167.189 \text{ [m]} + 89.88 (9.16) + \frac{1}{2} (-9.81) (9.16^2)$$

$$X_{\text{apogee}} \approx 411.7 \text{ [m]} \approx 412 \text{ [m]}$$

$$\textcircled{a} \text{ sampling rate } \geq 45 \text{ Hz} \rightarrow t_{\text{apogee}} = t_{\text{burn}} + t_{\text{drift}} = 1.85 + 9.16$$

$$t_{\text{apogee}} = 11.015 \text{ [s]} \Rightarrow 11.01 \cdot 45 \text{ [Hz]} \approx 495 \text{ samples}$$

Power Capacity Calculation

- assuming $\approx 5 \text{ [m/s]}$ descent rate,

$$\Delta h = V_{\text{descent}} \cdot t \rightarrow 412 \text{ [m]} = 5 \text{ [m/s]} \cdot t_{\text{descent}}$$

$$\therefore t_{\text{descent}} = 82.4 \text{ [s]} \rightarrow t_{\text{total}} = 10 + 1.85 + 9.16 + 82.4$$

$$t_{\text{total}} \approx 104 \text{ sec} \rightarrow \text{Capacity req: } \geq 2 \text{ min operation}$$